Excluding the next three first shared labels

Human Compression project: L044 continuation

Retain the full critical six-point-head hypotheses and all complete predecessor proofs. Thus $n = 2r \geq 18$, actual ambient IP holds, and the critical tail is the natural alternating group $R = G_{\geq 3} = \mathrm{Alt}((\Omega \setminus D) \cup \{6\})$. Write

$$\tau = \rho_3 = (6\ b)\delta, \qquad \sigma = \rho_4 = (b\ x)\epsilon, \qquad W = D \cup \{b, x\}.$$

The accepted pure-overlap argument makes $|W| = 8$ and makes δ, ϵ odd involutions fixing W pointwise. The complete prefix-kernel proof supplies the strictly increasing head orbits C_k, starting with $|C_3| = 7$, $|C_4| = 8$, and the pure subgroup $\mathrm{Alt}(C_k) \leq P_k = G_{\leq k}$. Every later unique cut has omission orbits C_{j-1} and its complement, with all shared labels smaller than j.

Lemma 1 (The exterior support and its four-point orbit)**.** *The support of $J = \langle \delta, \epsilon \rangle$ has at least six points, so P_4 has at least fourteen moved points. If δ is a single transposition $(u\ v)$, the group $J = \langle \delta, \epsilon \rangle$ has a four-point orbit $U = \{u, v, u^\epsilon, v^\epsilon\}$, disjoint from W.*

Proof. Let $m = |\delta\epsilon|$. The pure three-cycle on $\{6, b, x\}$ belongs to $P_4 \cap R = \langle \tau, \sigma \rangle$, by actual IP and the natural alternating groups on W and on the critical tail. A nonidentity odd-order element in this group generated by two involutions belongs to its cyclic rotation subgroup. Since $\tau\sigma$ has a three-cycle on $\{6, b, x\}$ and restriction $\delta\epsilon$ outside W, a power producing that pure three-cycle must have exponent divisible by m and not divisible by 3. Hence $3 \nmid m$.

The involutions δ and ϵ do not commute. If they did, δ would centralize both τ and σ. It also centralizes $G_{\geq 5}$: those generators fix $6, b$ and commute with τ. Thus it would centralize the transitive critical tail R while fixing 6, and would be the identity, contrary to its oddness.

This argument applies to every odd nonidentity δ. Therefore $m \geq 4$ as well as $3 \nmid m$. On at most five points every odd involution is a single transposition, and the product of two transpositions has order at most three. Thus J moves at least six points, all outside W. The support of P_4 is W together with that exterior support, proving the general assertion. For the remaining assertion assume $\delta = (u\ v)$.

If ϵ preserved $\{u, v\}$, it would commute with δ. If it moved just one of these points outside the pair, $\delta\epsilon$ would have a three-cycle on that point, its image and the fixed endpoint of the pair. All other cycles would have length at most two; its order would be divisible by three, again impossible. Therefore ϵ moves both points outside the pair, to distinct points u', v'. On $U = \{u, v, u', v'\}$ the two generators are $(u\ v)$ and $(u\ u')(v\ v')$, which are transitive together. This set is invariant under both involutions and is disjoint from W, proving the assertion. □

Theorem 2 (Labels five and six are excluded as well)**.** *At a unique eligible cut of index at least five, the first shared label cannot be five or six.*

Proof. Let g be the first shared label at such a cut j. All shared labels are smaller than j, and the prefix P_{g-1} fixes the other omission orbit: every earlier mark acts on the head orbit and is unshared. In the reversed string its index is $h = r - 1 - g$, the last shared index. The full interior or endpoint support bound gives

$$|\operatorname{supp}(P_{g-1})| \leq n - (2h + 1) = 2g + 1.$$

For $g \geq 5$ this prefix contains P_4 and thus moves at least fourteen points by Lemma 1. For $g = 5, 6$ the displayed upper bound is respectively eleven or thirteen, a contradiction. □

Theorem 3. *At a unique eligible cut of index $j \geq 5$, the first shared label cannot be* 4.

Proof. Suppose the first shared label is 4, and write $C = C_{j-1}$, $A = \Omega \setminus C$ for the omission orbits. The prefix P_3 fixes A pointwise, since its head labels act on C and none of its four labels is shared. Its support is the disjoint union of $D \cup \{b\}$ and $\mathrm{supp}(\delta)$.

Reverse the entire string. The last shared index is now $h = r - 5$, and the tail past h is the actual P_3. The full reversed support bound, interior or endpoint as appropriate, gives

$$n - |\,\mathrm{supp}(P_3)| \geq 2h + 1 = 2r - 9.$$

Consequently $|\,\mathrm{supp}(P_3)| \leq 9$. The odd nonidentity δ fixes $D \cup \{b\}$, so its support has at least two points. Thus equality is forced, and δ is a single transposition.

First suppose $j < r - 1$. The reversed crossing has interior index $i = r - 1 - j \geq 1$. Its earlier subgroup $G_{>j}$ fixes C pointwise, by the accepted actual prefix-orbit lemma. The fixed set of the tail past h has size exactly $2h + 1$. If that earlier subgroup is transitive on A, the complete transitive-side support theorem gives $h = i+1$. If it is intransitive on A, the complete intransitive-side support theorem gives the same conclusion at equality. All their hypotheses have the actual reversed meanings: the second orbit C is nontrivial, the earlier subgroup fixes it, and the tail past the last shared label fixes A. Hence in either case $j = 5$. But then $C = C_4$ has size eight, whereas P_3, supported inside C, has nine moved points. This contradiction excludes every interior cut in the statement.

It remains to consider $j = r - 1$. The group P_{r-2} is transitive on A, fixes its complement C setwise, and its first four generators fix A pointwise. Thus the restrictions of $\rho_4, \ldots, \rho_{r-2}$ generate a transitive string action on A. There are $d = r - 5$ such labels, and the restriction of ρ_4 is nonidentity by the first-shared hypothesis. The endpoint $a \in A$ of the unique crossing is fixed by every generator of index at most $r-3$. The complete fixed-prefix/no-gap argument in the reversed order therefore gives

$$|A| \geq d + 1 = r - 4. \tag{1}$$

Explicitly, a shortest path from a to a point moved by ρ_4 uses every intervening label $r-2, \ldots, 5$; appending its 4-edge gives d edges and $d+1$ distinct vertices. The path exists by transitivity and nontriviality of the last restriction. This is the minimum-length part of the anchored-path proof, with no bound on $|A|$ assumed.

Lemma 1 now supplies the four-point P_4-orbit U outside $C_4 = W$. Both points moved by δ belong to C, because P_3 fixes A. Since $P_4 \leq P_{r-2}$, the whole orbit U is contained in C. Let $k \geq 5$ be the first index for which C_k meets U. The previous head orbit is disjoint from U, and invariance under $P_4 \leq P_k$ makes C_k contain all four points. Thus this one strict increment has size at least four. All other increments have size at least one. Starting from $|C_3| = 7$, there are $r-5$ increments up to C_{r-2}, so

$$|C| \geq 7 + (r - 5) + 3 = r + 5. \tag{2}$$

Equations (1)–(2) give $n = |A| + |C| \geq 2r + 1$, contrary to $n = 2r$. The terminal case is excluded as well. □

Attribution and remaining scope. The numerical outside-support bounds and their exact interior equality statements remain the complete acquired proofs of the source L044 Propositions 5.20, 5.22 and 5.23, physical pages 30–33. The prefix kernels and literal omission-orbit formulas also remain paid, with the initial parity-fibre contribution attributed there to the root collaborator. The six-point lower bound for the exterior support repeats the initial argument of the acquired opposite-end proof; it does not use the second head. The literal four-point exterior orbit and its cost in the strict prefix growth are the present argument. No IP is assigned to an exterior image. Together with the independently reviewed index-three exclusions, any shared later unique cut now has first shared label at least seven. The other shared cuts, opposite heads, terminal singleton and full source contracts remain separate. No whole-original, conjecture, numerical or formalization credit is claimed.